

ERRORS IN C

Types Of Errors

1) Compile Time Errors

- Pre-processor stage error
- Translator stage error
- Linker stage error

2) Run Time Errors

- Segmentation fault
- Floating point exception
- Bus error

Compile Time Errors

1) Pre-processor stage error:

When pre-processor stage generates error ?

- **Pre-processor Directive's spelling mistake.**

EX: `#includ<stdio.h> //include`

- **Wrong header file name.**

EX: `#include<stio.h> //stdio.h`

2) Translator stage error:

When translator stage generates error ?

- **Because of syntactical errors Translator stage generates error.**

EX: int a,b //missing ;(semicolon)

3) Linker stage error:

When linker stage generates error ?

- **linker is to link the function calls with function definitions.**
If at this stage function definition not found, linker will generate error.

EX: `print("Vector India Pvt Ltd");`
`//library(predefined) function is printf`

Run Time Errors

1) Segmentation fault:

- segmentation fault is nothing but unauthorised memory access. when executable file is copied to RAM for execution, it is allotted with some memory space. If program tries to access beyond those memory limits, segmentation fault error occurs.

EX: `scanf(“%d”,a);` `// & (reference operator) is not used with variable a`

2) Floating point exception:

- This error occurs when integer value is divided by zero while execution.

3) Bus error:

- Bus error is low level hardware related error. When program tries to access the memory location whose address is not present in the RAM, bus error generates.